Ex.No.14

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WRITE A CODE USING RAW SOCKETS TO IMPLEMENT PACKET SNIFFING

AIM:

To Implement a simple packet sniffer in Java that captures and displays network packets.

ALGORITHM:

1) Start the program
2) Create a RAW socket using socket(AF_PACKET, SOCK_RAW)
3) Bind the socket to the required network interface.
4) Continuously receive packet from the network
5) Extract the Ethernet header from each packet.
6) Identify the protocol type, such as IPv4 ARP or IPv6
7) IPv4 packets, extract the source and destination IP address
8) Display the packet information
9) Repeat 6 until the user terminates the program
10) Close the socket.

PROGRAM:

```
import socket
import struct
raw_socket = socket.socket ( socket.AF_PACKET , socket.SOCK_RAW,
                             socket.ntohs(3));

print (" Packet Sniffer started ...")
print ("waiting for packets...\n")
```

```
try:
    while True:
        packet, address = raw_socket.recvfrom(65535)
        ethernet_header = packet[0:14]
        dest_mac, src_mac, protocol = struct.unpack("!6s6sH",
                                      ethernet_header)
        dest_mac = ":".join("%.02x" % byte for byte in dest_mac)
        src_mac = ":".join("%.02x" % byte for byte in src_mac)
        print("=========")
        print("SourceMAC:", src_mac)
        print("DestinationMAC:", dest_mac)
        print("Protocol:", hex(protocol))
        if protocol == 0x0800:
            ip_header = packet[14:34]
            version_ihl, tos, total_length, identification, flags_fragment,
            ttl, proto, checksum, src_ip, dest_ip = struct.
                unpack("!BBHHHBBHII", ip_header)
            src_ip = socket.inet_ntoa(src_ip)
            dest_ip = socket.inet_ntoa(dest_ip)
        elif protocol == 0x0806:
            print("packetType: ARP")
        elif protocol == 0x86dd:
            print("packetType: IPv6")
```

SAMPLE OUTPUT:

packet sniffer started

waiting for packets

Source MAC: 3C:52:82:11:22:33

Destination MAC: ff:ff:ff:ff:ff:ff

protocol: 0x306

packet type: ARP

= = = = = = = =

Source MAC: 3C:52:82:11:22:33

Destination MAC: 84:16:64:44:55:66

protocol: 0x800

Source IP: 192.168.1.10

Destination IP: 192.168.1.1

TTL: 64

IP protocol: 6

= = = = = = = =

```
        else:
            print("packet Type: Other")
    except KeyboardInterrupt:
        print("\n Packet Sniffing Stopped.")
    finally:
        raw_socket.close()
```

RESULT:

Thus, a packet_sniffing programusing RAW Socket was successfully implemented. the program captures network packets & display important information such as source MAC address, destination MAC address, source IP address destination IP address and TTL.